Intel x86 Processors

- **Totally dominate laptop/desktop/server market**

- **Evolutionary design**
  - Backwards compatible up until 8086, introduced in 1978
  - Added more features as time goes on

- **Complex instruction set computer (CISC)**
  - Many different instructions with many different formats
    - But, only small subset encountered with Linux programs
  - Hard to match performance of Reduced Instruction Set Computers (RISC)
  - But, Intel has done just that!
    - In terms of speed. Less so for low power.

Intel x86 Evolution: Milestones

Name	*Date*	*Transistors*	*MHz*
8086	**1978**	**29K**	**5-10**

- First 16-bit Intel processor. Basis for IBM PC & DOS
- 1MB address space

386	**1985**	**275K**	**16-33**

- First 32 bit Intel processor , referred to as IA32
- Added "flat addressing", capable of running Unix

Pentium 4F	**2004**	**125M**	**2800-3800**

- First 64-bit Intel processor, referred to as x86-64

Core 2	**2006**	**291M**	**1060-3500**

- First multi-core Intel processor

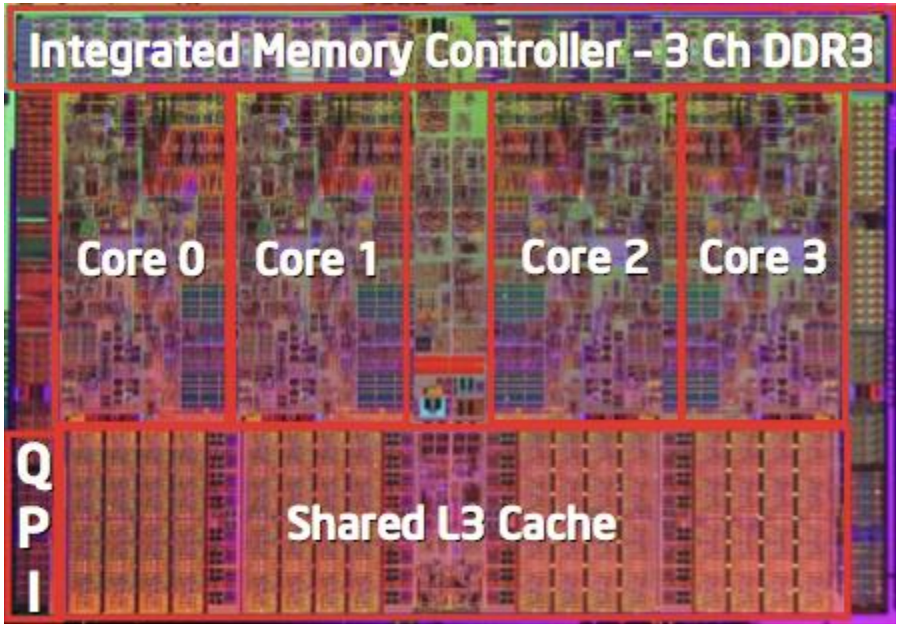
Core i7	**2008**	**731M**	**1700-3900**

- Four cores (our shark machines)

Intel x86 Processors, cont.

Machine Evolution

386	1985	0.3M
Pentium	1993	3.1M
Pentium/MMX	1997	4.5M
PentiumPro	1995	6.5M
Pentium III	1999	8.2M
Pentium 4	2001	42M
Core 2 Duo	2006	291M
Core i7	2008	731M



Added Features

- Instructions to support multimedia operations
- Instructions to enable more efficient conditional operations
- Transition from 32 bits to 64 bits
- More cores

x86 Clones: Advanced Micro Devices (AMD)

- **Historically**
  - AMD has followed just behind Intel
  - A little bit slower, a lot cheaper
- **Then**
  - Recruited top circuit designers from Digital Equipment Corp. and other downward trending companies
  - Built Opteron: tough competitor to Pentium 4
  - Developed x86-64, their own extension to 64 bits

Intel's 64-Bit

- **Intel Attempted Radical Shift from IA32 to IA64**
  - Totally different architecture (Itanium)
  - Executes IA32 code only as legacy
  - Performance disappointing
- **AMD Stepped in with Evolutionary Solution**
  - x86-64 (now called "AMD64")
- **Intel Felt Obligated to Focus on IA64**
  - Hard to admit mistake or that AMD is better
- **2004: Intel Announces EM64T extension to IA32**
  - Extended Memory 64-bit Technology
  - Almost identical to x86-64!
- **All but low-end x86 processors support x86-64**
  - But, lots of code still runs in 32-bit mode

Our Coverage

- **IA32**
  - The traditional x86
  - `shark> gcc -m32 hello.c`
- **x86-64**
  - The emerging standard
  - `shark> gcc hello.c`
  - `shark> gcc -m64 hello.c`

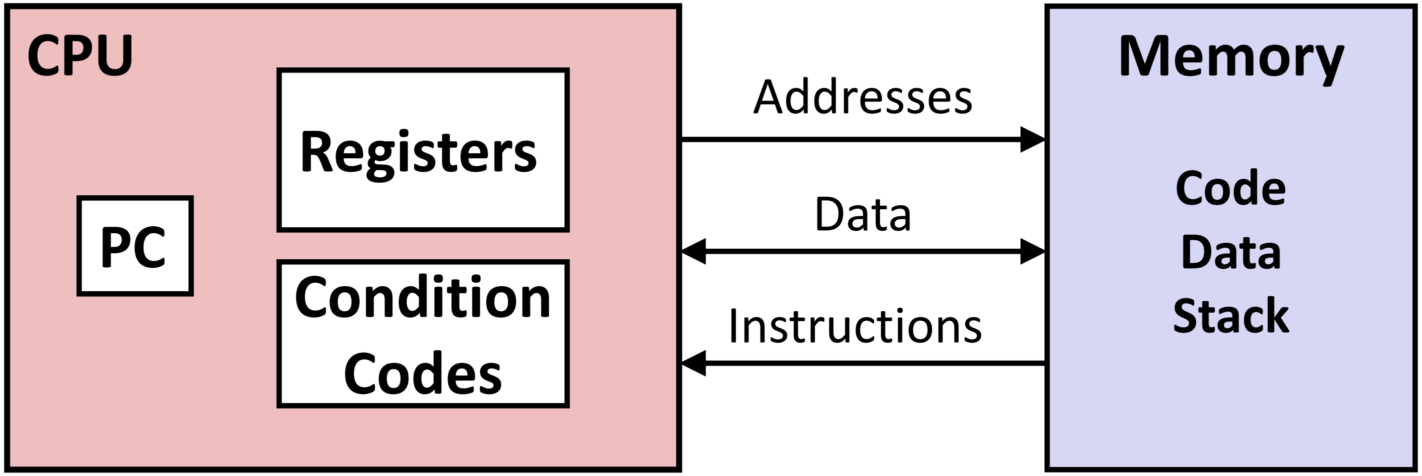
- **Presentation**
  - Book presents IA32 in Sections 3.1—3.12
  - Covers x86-64 in 3.13
  - We will cover both simultaneously
  - Some labs will be based on x86-64, others on IA32

Definitions

- **Architecture: (also ISA: instruction set architecture) The parts of a processor design that one needs to understand to write assembly code.**
  - Examples: instruction set specification, registers.
- **Microarchitecture: Implementation of the architecture.**
  - Examples: cache sizes and core frequency.

- **Example ISAs (Intel): x86, IA**

Assembly Programmer's View

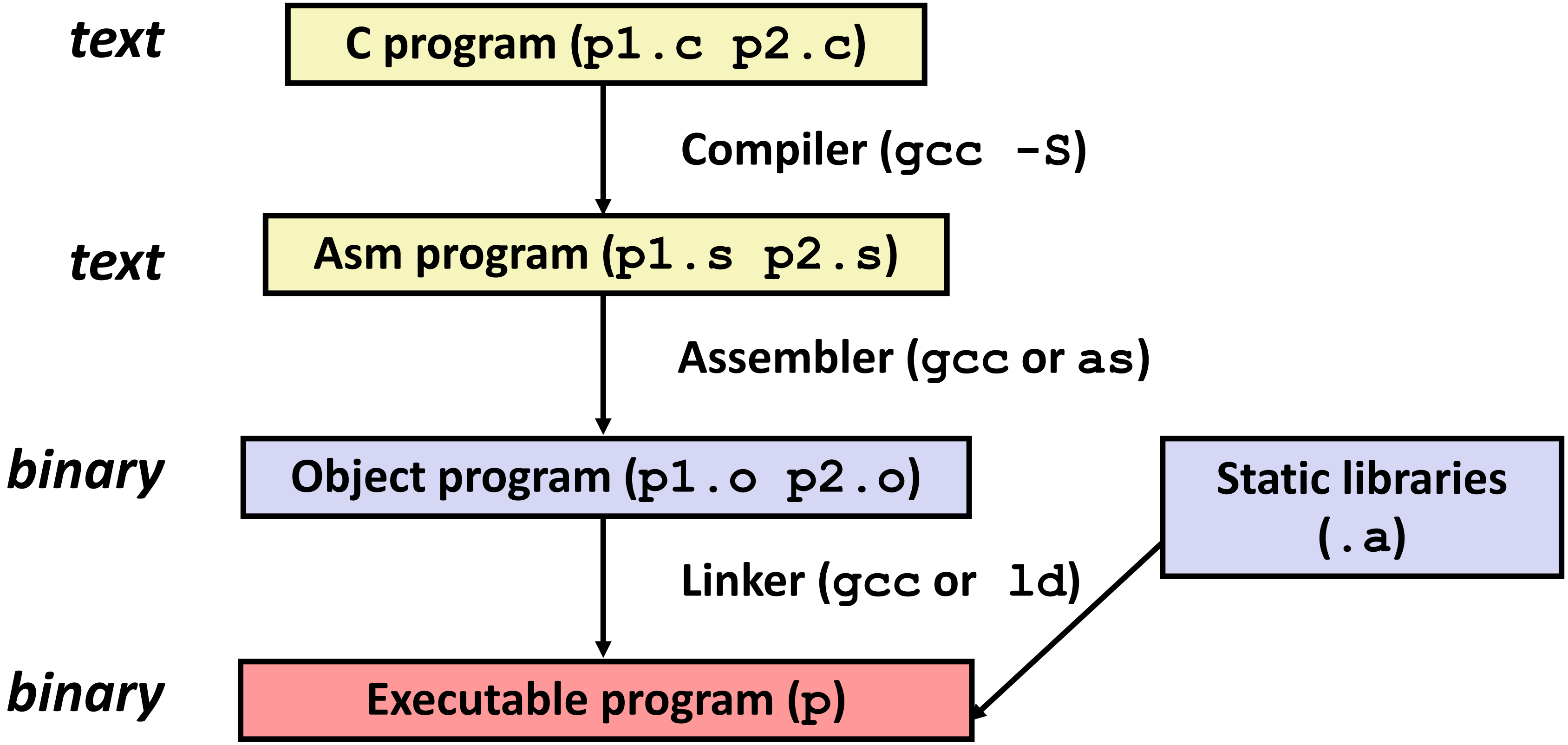


Programmer-Visible State

- **PC: Program counter**
  - Address of next instruction
  - Called "EIP" (IA32) or "RIP" (x86-64)
- **Register file**
  - Heavily used program data
- **Condition codes**
  - Store status information about most recent arithmetic operation
  - Used for conditional branching
- **Memory**
  - Byte addressable array
  - Code and user data
  - Stack to support procedures

Turning C into Object Code

- Code in files `p1.c p2.c`
- Compile with command: `gcc -Og p1.c p2.c -o p`
  - Use basic optimizations (`-Og`)
  - Put resulting binary in file `p`



Compiling Into Assembly

C Code

```
int sum(int x, int y)
{
  int t = x+y;
  return t;
}
```

Generated IA32 Assembly

```
sum:
   pushl %ebp
   movl %esp,%ebp
   movl 12(%ebp),%eax
   addl 8(%ebp),%eax
   popl %ebp
   ret
```

Obtain with command

```
/usr/local/bin/gcc –O1 -S code.c
```

Produces file `code.s`

Assembly Characteristics: Data Types

- **"Integer" data of 1, 2, or 4 bytes**
  - Data values
  - Addresses (untyped pointers)

- **Floating point data of 4, 8, or 10 bytes**

- **No aggregate types such as arrays or structures**
  - Just contiguously allocated bytes in memory

Assembly Characteristics: Operations

- **Perform arithmetic function on register or memory data**

- **Transfer data between memory and register**
  - Load data from memory into register
  - Store register data into memory

- **Transfer control**
  - Unconditional jumps to/from procedures
  - Conditional branches

Object Code

Code for `sum`

```
0x401040 <sum>:
    0x55
    0x89
    0xe5
    0x8b
    0x45
    0x0c
    0x03
    0x45
    0x08
    0x5d
    0xc3
```

- **Total of 11 bytes**
- **Each instruction 1, 2, or 3 bytes**
- **Starts at address `0x401040`**

- **Assembler**
  - Translates `.s` into `.o`
  - Binary encoding of each instruction
  - Nearly-complete image of executable code
  - Missing linkages between code in different files
- **Linker**
  - Resolves references between files
  - Combines with static run-time libraries
    - E.g., code for **`malloc`**, **`printf`**
  - Some libraries are *dynamically linked*
    - Linking occurs when program begins execution

Machine Instruction Example

```
int t = x+y;
```

```
addl 8(%ebp),%eax
```

Similar to expression:

```
x += y
```

More precisely:

```
int eax;
int *ebp;
eax += ebp[2]
```

```
0x80483ca:   03 45 08
```

- **C Code**
  - Add two signed integers
- **Assembly**
  - Add 2 4-byte integers
    - "Long" words in GCC parlance
    - Same instruction whether signed or unsigned
  - Operands:

    x: Register `%eax`

    y: Memory `M[%ebp+8]`

    t: Register `%eax`

    – Return function value in `%eax`
- **Object Code**
  - 3-byte instruction
  - Stored at address `0x80483ca`

Disassembling Object Code

Disassembled

```
080483c4 <sum>:
 80483c4:   55          push   %ebp
 80483c5:   89 e5       mov    %esp,%ebp
 80483c7:   8b 45 0c    mov    0xc(%ebp),%eax
 80483ca:   03 45 08    add    0x8(%ebp),%eax
 80483cd:   5d          pop    %ebp
 80483ce:   c3          ret
```

- **Disassembler**

  `objdump -d p`

  - Useful tool for examining object code
  - Analyzes bit pattern of series of instructions
  - Produces approximate rendition of assembly code
  - Can be run on either `a.out` (complete executable) or `.o` file

Alternate Disassembly

Object

```
0x401040:
    0x55
    0x89
    0xe5
    0x8b
    0x45
    0x0c
    0x03
    0x45
    0x08
    0x5d
    0xc3
```

Disassembled

```
Dump of assembler code for function sum:
0x080483c4 <sum+0>:     push   %ebp
0x080483c5 <sum+1>:     mov    %esp,%ebp
0x080483c7 <sum+3>:     mov    0xc(%ebp),%eax
0x080483ca <sum+6>:     add    0x8(%ebp),%eax
0x080483cd <sum+9>:     pop    %ebp
0x080483ce <sum+10>:    ret
```

- **Within gdb Debugger**

  `gdb p`

  `disassemble sum`

  - Disassemble procedure

  `x/11xb sum`

  - Examine the 11 bytes starting at `sum`

What Can be Disassembled?

```
% objdump -d WINWORD.EXE

WINWORD.EXE:   file format pei-i386

No symbols in "WINWORD.EXE".
Disassembly of section .text:

30001000 <.text>:
30001000:  55              push   %ebp
30001001:  8b ec           mov    %esp,%ebp
30001003:  6a ff           push   $0xffffffff
30001005:  68 90 10 00 30  push   $0x30001090
3000100a:  68 91 dc 4c 30  push   $0x304cdc91
```

- Anything that can be interpreted as executable code
- Disassembler examines bytes and reconstructs assembly source

Integer Registers (IA32)

general purpose

				Origin (mostly obsolete)
`%eax`	`%ax`	`%ah`	`%al`	*accumulate*
`%ecx`	`%cx`	`%ch`	`%cl`	*counter*
`%edx`	`%dx`	`%dh`	`%dl`	*data*
`%ebx`	`%bx`	`%bh`	`%bl`	*base*
`%esi`	`%si`			*source index*
`%edi`	`%di`			*destination index*
`%esp`	`%sp`			***stack pointer***
`%ebp`	`%bp`			***base pointer***

16-bit virtual registers (backwards compatibility)

Moving Data: IA32

- **Moving Data**

  `movl` ***Source*, *Dest***:

- **Operand Types**
  - ***Immediate:*** Constant integer data
    - Example: **`$0x400`, `$-533`**
    - Like C constant, but prefixed with **`'$'`**
    - Encoded with 1, 2, or 4 bytes
  - ***Register:*** One of 8 integer registers
    - Example: **`%eax, %edx`**
    - But **`%esp`** and **`%ebp`** reserved for special use
    - Others have special uses for particular instructions
  - ***Memory:*** 4 consecutive bytes of memory at address given by register
    - Simplest example: **`(%eax)`**
    - Various other "address modes"

`%eax`
`%ecx`
`%edx`
`%ebx`
`%esi`
`%edi`
`%esp`
`%ebp`

`movl` Operand Combinations

	Source	Dest	Src,Dest	C Analog
`movl`	*Imm*	*Reg*	`movl $0x4,%eax`	`temp = 0x4;`
		Mem	`movl $-147,(%eax)`	`*p = -147;`
	Reg	*Reg*	`movl %eax,%edx`	`temp2 = temp1;`
		Mem	`movl %eax,(%edx)`	`*p = temp;`
	Mem	*Reg*	`movl (%eax),%edx`	`temp = *p;`

Cannot do memory-memory transfer with a single instruction